Alejandro Rosales

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Qualifications Summary

Computational Biologist/Data Engineer Intern with expertise in driving efficiency, data optimization, automation, and agile methodologies.

Software / Data Engineering

- Develop and maintain software applications and data pipelines for various business needs.
- Build data architectures for efficient storage, retrieval, and processing of large datasets.

Web Applications Programming

- Design responsive, interactive web applications meeting user and business objectives.
- Proficient in AWS, SQL, RESTful APIs, JavaScript, HTML/CSS, and Python.

Project Management

- Spearhead projects from conception to completion, ensuring on-time, on-budget delivery.
- Monitor project progress, track milestones, and manage risks effectively.

Key Strengths

- Drive enhancements in current and established computing systems.
- Leverage cloud computing and programming languages to craft functional, dynamic solutions.

Career Experience

Brightseed, South San Francisco, CA **2022 – 2023**

Computational Biologist/Data Engineer Intern

Reduced data access time for Brightseed's proprietary information from weeks to hours by integrating Neo4j graph database and internal API solutions, conducting data analysis with the senior data engineer. Cut AI algorithm setup from days to minutes using Docker containerization. Applied Agile methodology, maintained code documentation, version control via Git, unit testing, and team repository management on GitHub.

- Achieved 10x efficiency gain in generating core AI pipeline data packages at Brightseed through automation with the senior data engineer.
- Reduced computation time for 2.5B AI predictions from three weeks to three days via program profiling, resource optimization, and parallel processing across the Brightseed server.
- Boosted core AI algorithm efficiency 30% by building and deploying front-end and back-end components for an internal website.
- Delivered projects using AWS (S3, EC2, ECS, ALB, VNC, Lambda) and a broad tech stack: Python, PHP, Angular, Docker, REST API, JavaScript, JSON, TypeScript, jQuery, Ajax, SQL, NoSQL, MongoDB, Neo4j.
- Presented three projects company-wide, covering implications, use cases, and methodologies under the director of computation.

The Coder School, Westerville, OH **2019 – 2021**

Code Coach

Developed students' programming skills with emphasis on runtime and space optimization, supporting objectives in website design, language acquisition, and machine learning through targeted coaching.

- Designed standardized Python curriculum for summer camps, selecting teaching materials and course content per school requirements.

Education

Master's Student in Computer Science | Focus in Quantum Machine Learning, Ohio University, Athens, OH

- Relevant Coursework: Quantum Information Processing, Hardware for Deep Learning, Advanced Algorithms
- Research: Quantum Convolutional Neural Networks (QCNNs), Quantum Error Correction, Quantum Low-Density Parity-Check (qLDPC) Codes
- Related Projects (in progress): IEEE Quantum Week Technical Paper on novel error corrected QCNN methodology integrating qLDPC codes, Tutorial Journal Paper on QCNNs. MS Thesis

Bachelor of Science in Computer Science | Minor in Philosophy, Miami University, Oxford, OH

- Relevant Coursework: Advanced Algorithms, Artificial Intelligence, Website Design, App Design
- Academic Affiliations: Curriculum Executive, Miami University Python | Miami University Programming Club